

# JIAYI WANG

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## RESEARCH INTERESTS

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- Computer architecture for AI and general-purpose workloads (CPU/GPGPU/accelerators).
- Reconfigurable and spatial/dataflow computing architectures (FPGA/CGRA).
- Memory systems, cross-stack co-design, and multimodal AI systems for real-world applications.

## EDUCATION

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**University of Washington, Seattle** Sep 2023 – June 2028 (expected)  
Ph.D. in Computer Engineering  
GPA: 4.00  
Focus: Computer architecture, reconfigurable AI hardware architecture, AI systems, digital VLSI  
Advisor: Prof. Ang Li

**University of Washington, Seattle** Sep 2020 – Aug 2022  
M.S. in Electrical Engineering  
GPA: 4.00  
Focus: Computer architecture, digital VLSI

**Shanghai Jiao Tong University** Sep 2016 – Jun 2020  
B.S. in Electrical Engineering  
GPA: 3.70  
Focus: Power electronics, embedded systems

**University of California, Los Angeles** Summer 2019  
Summer School, Computer Science  
GPA: 4.00

## FIRST-AUTHOR PUBLICATIONS AND WORKSHOPS

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- [1] **Jiayi Wang**, Ang Da Lu, Zhichen Zeng, and Ang Li,  
“*DICE: Enabling Efficient General-Purpose SIMT Execution with Statically Scheduled Coarse-Grained Reconfigurable Arrays*,”  
in **The 53rd Annual International Symposium on Computer Architecture (ISCA 2026)**, Raleigh, NC, USA, Jun. 27–Jul. 1, 2026. (Acceptance rate: 19.1%.) [[arXiv:2605.05496](https://arxiv.org/abs/2605.05496)]
- [2] **Jiayi Wang**, Kearnan Bishop, and Ang Li,  
“*MFSA: A Multi-Format Systolic Array with Native Micro Scaling Support*,”  
in **The 44th IEEE International Conference on Computer Design (ICCD 2026)**, Hong Kong, Nov. 16–18, 2026. (Acceptance rate: 26%.)
- [3] **Jiayi Wang**, Maohua Nie, Sin-Chen Lin, C.-J. Richard Shi, and Ang Li,  
“*TransDot: An Area-Efficient Reconfigurable Floating-Point Unit for Trans-Precision Dot-Product Accumulation for FPGA AI Engines*,”  
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**, Atlanta, GA, USA, May 13–16, 2026. (Acceptance rate: 24.6%.) [[IEEE Xplore](https://ieeexplore.org/)]
- [4] **Jiayi Wang**, Ang Da Lu, and Ang Li,  
“*DICE: Efficient Thread-Pipelined SIMT Execution on a Statically Scheduled Reconfigurable Spatial Fabric*,”  
**Presented at 18th Workshop on General Purpose Processing Using GPU (GPGPU 2026)**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026.

- [5] **Jiayi Wang**, Chenyi Wang, and Ang Li,  
“*piPE-SA: Enabling Deeply Pipelined Processing Elements in Systolic Arrays*,”  
in **Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop**, held in  
conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.
- [6] **Jiayi Wang**, Yiwei Yang, Jiajin Wu, Zheng Zeng, Di Lu, and Lian Lian,  
“*Hybrid Aerial-Aquatic Vehicle for Large-Scale High-Spatial-Resolution Marine Observation*,”  
in **OCEANS 2019 Marseille**, Jun. 2019, pp. 1–7. doi: [10.1109/OCEANSE.2019.8867402](https://doi.org/10.1109/OCEANSE.2019.8867402).

## COLLABORATIVE PUBLICATIONS

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- [7] Jingqun Zhang, Maohua Nie, **Jiayi Wang**, Weihang Li, Rishi Sappidi, Shwet Chitnis, and Ang Li,  
“*CacheFlex: Direct Software-Managed Access to Higher-Level Cache for Scalable Vector Support*,”  
in **The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)**. (Acceptance rate: 25.3%.)
- [8] Ryan H. Lee, Kate L. Tseng, Te Min Yu, Andrew Pan, **Jiayi Wang**, James Rosenthal, Kevin J. Ho, Ang Li, and Matthew Reynolds,  
“*All-Digital Bluetooth Low Energy (BLE) Backscatter ASIC Using Standard I/O Pad Drivers in 180 nm CMOS*,”  
in **Proceedings of the 2026 IEEE International Conference on RFID (RFID 2026)**.
- [9] Jingqun Zhang, Weihang Li, Maohua Nie, Yung-Jen Cheng, **Jiayi Wang**, Shwet Chitnis, and Ang Li,  
“*CacheFlex: Explicitly Control What You Need in Your Cache*,”  
**Presented at Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.
- [10] Ang Da Lu, **Jiayi Wang**, and Ang Li,  
“*STEP: Spatially Threaded Execution Pipeline*,”  
in **Proceedings of the 6th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE 2026)**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026. [\[PDF\]](#)
- [11] Shwet Chitnis, Fergus Xu, Ayush Kulkarni, **Jiayi Wang**, Jingqun Zhang, Arjun Raje, and Ang Li,  
“*Precision-Aware Communication in CGRAs*,”  
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**.
- [12] Shwet Chitnis, **Jiayi Wang**, Fergus Xu, Ayush Kulkarni, Rampranav Navendran, Juwon Jun, Jingqun Zhang, and Ang Li,  
“*DORA: Open-Source Infrastructure for Prototyping Reconfigurable Fabrics*,”  
**Poster at Workshop on Open-Source Computer Architecture Research (OSCAR 2026)**, held in conjunction with ISCA 2026, Raleigh, NC, USA, Jun. 2026.

## PATENTS

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- [1] Ang Li and **Jiayi Wang**,  
“*DICE: Enabling Efficient General-Purpose SIMT Execution with Coarse-Grained Reconfigurable Arrays*,”  
**U.S. Provisional Patent Application 64/012,051**, filed Mar. 20, 2026.
- [2] Zheng Zeng, **Jiayi Wang**, Yiwei Yang, Jiajin Wu, Lian Lian, and Di Lu,  
“*An Aerial-Aquatic Vehicle Combining a Fixed-Wing Aircraft and a Glider*,”  
**Chinese Invention Patent CN 109204812 B**, granted Nov. 17, 2020.

## SELECTED RESEARCH EXPERIENCE

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**Reconfigurable AI Hardware and Accelerator Microarchitecture Research**  
*UW P<sup>N</sup> Computer Engineering Lab (PI: Prof. Ang Li)*

*Aug 2025 – Present*

- Building multi-format, reconfigurable accelerator building blocks, from the ALU up to the systolic array, for AI workloads that mix FP32/BF16/FP8/FP4, block-scaled (MX), and integer formats and for the small, non-square GEMMs of LLM serving.
- **ALU level – TransDot:** a reconfigurable floating-point unit covering FP4/FP8/FP16/FP32 FMA, subword SIMD, and FP16/FP32 dot-product accumulation, replacing per-format FPUs in FPGA AI engines.
- **Systolic-array level – MFSA:** a weight-stationary multi-format systolic array with a shared dot-product datapath from FP32 to FP4/INT4 and online block-scale rebasing that folds MX/NVFP4 block scales into the running partial sum, cutting block-scaling overhead by roughly two thirds in area and energy.
- **Systolic-array level – piPE-SA:** a pipeline-aware systolic array that keeps weight-/output-stationary timing correct with multi-stage pipelined PEs, making PE pipeline depth a design axis alongside array size and dataflow.
- **Fracturable, skew-free systolic array (ongoing):** a twisted-torus fabric with an orthogonal wavefront that removes the diagonal skew of classical arrays and can be partitioned at runtime into independently scheduled strips.
- Related publications: [2], [3], [5]; one manuscript under review.

### High-Efficiency CGRA–GPU/CPU Integration Research

*UW P<sup>N</sup> Computer Engineering Lab (PI: Prof. Ang Li)*

*Oct 2024 – Present*

- Fusing statically scheduled CGRAs with SIMT execution to keep the CUDA programming model while removing register-file and per-instruction control overhead, on both GPUs and CPUs.
- **DICE (GPU side):** replaces the SM’s SIMD backend with statically scheduled CGRAs, with a Temporal Memory Coalescing Unit and swizzled thread-bank mapping to remove register-file bank conflicts; **1.90× energy efficiency, 42% lower power, and 1.16× performance** vs. NVIDIA Turing SMs. Filed as a U.S. provisional patent.
- **Compiler and modeling:** built a *p-graph* compiler and, with STEP, the spatially threaded execution pipeline that maps SIMT kernels onto the fabric; extended GPGPU-Sim/GPUWattch for performance and power evaluation on GPU-Rodinia.
- **CPU side (ongoing):** a core-private SIMT engine on the same reconfigurable substrate, attached to each CPU core’s cache hierarchy with an in-memory thread-launch structure, so fine-grained irregular kernels run without driver launches or explicit data movement.
- Led an 8-member RTL team to build the RTL implementation and FPGA prototype for architectural validation.
- Related publications: [1], [4], [10]; one manuscript under review.

### Coarse-Grained Reconfigurable Architecture Framework Research

*UW P<sup>N</sup> Computer Engineering Lab (PI: Prof. Ang Li)*

*Mar 2024 – Present*

- Building open-source infrastructure that makes reconfigurable fabrics fast to generate, evaluate, and prototype, instead of rewriting RTL for every processing-element, routing, or control variant.
- **DORA:** a Python-based, highly parameterized CGRA RTL generator with customizable processing elements, routing networks, and control flow, used across the lab’s CGRA projects for rapid architecture exploration and FPGA prototyping.
- **Precision-aware communication:** studied how operand precision can shape data movement and interconnect design in CGRAs.
- Related publications: [11], [12].

### GPU Multi-Tenant Power Management Research

*UW P<sup>N</sup> Computer Engineering Lab (PI: Prof. Ang Li)*

*2026 – Present*

- Datacenter GPUs run under power caps and are shared across tenants, yet partitions such as MIG share one device-wide cap and clock, so one tenant reaching the cap throttles every latency-critical neighbor.
- Characterized on H100/B200-class GPUs how a co-located tenant’s power draw inflates a prioritized neighbor’s time-to-first-token under power caps.

- Proposed making power an explicitly managed per-tenant resource: per-partition power metering, independent per-GPC voltage/frequency control, and a power arbiter that sets budgets by tenant priority and phase, updates them at sub-millisecond intervals, and lends unused budget.
- Evaluated with a serving simulator driven by real LLM traces and a power model calibrated on H100 and B200, restoring the prioritized tenant's latency to near its standalone value.

## AI Agent Development

*SABOL Vision Lab, Cultivate Learning (PI: Prof. Gail E. Joseph)*

*Feb 2026 – Present*

- **VISTA** (Video Interaction Scoring and Teaching Assistant): multimodal AI agents for early-childhood classrooms that turn classroom video into interaction-quality scores and real-time coaching for teachers.
- **VISTA-QEI**: a video-analysis agent that scores teacher-child interactions with the QEI framework and writes evidence-based coaching reports.
- **VISTA-RT**: a real-time coaching agent with live video/audio analysis, rolling context memory, and prioritized prompts delivered through wearable interfaces.
- Built the working demos and evaluation tools, and presented prototypes to AI and early-learning collaborators.

## High-Speed Interface Integration Research

*UW Silicon Systems Research Lab (PI: Prof. C.-J. Richard Shi)*

*Mar 2022 – Dec 2023*

- **CPHY**: a DDR/UCIe-compatible PHY with an LPDDR4/4x memory controller, taken from RTL to a TSMC N28 tape-out and integrated into a RISC-V SoC.
- Led RTL design, mixed-signal integration, and verification of the memory controller and PHY.
- Performed physical design in TSMC N28, including analog/mixed-signal integration.
- Integrated the controller and PHY with a RISC-V CPU and a deep-learning accelerator.

## INDUSTRY EXPERIENCE

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### NVIDIA Corporation

*Compute Performance Intern*

Summer 2026

- Working on GPU acceleration of Cadence EDA tools.

### Qualcomm Technologies

*Research Intern, GPGPU*

Jun 2024 – Sep 2024

- Investigated architectural optimizations in commercial GPGPU/NPU platforms to accelerate sparse matrix multiplication.
- Explored low-level sparse implementation flows in TensorRT, PyTorch, and NVIDIA CUTLASS using NVIDIA 2:4 Sparse Tensor Cores.
- Benchmarked and analyzed sparse Tensor Core performance across multiple models, including ResNet, MobileNet, and BERT.

### Orora Design Technologies

*Technical Staff*

Oct 2022 – Dec 2023

- Developed C++ automation tools for automatic SystemVerilog model generation of analog circuits.

### Intel Corporation

*Technical Enablement Engineer Intern*

Mar 2021 – Sep 2021

- Debugged and verified DDR and PCIe high-speed interfaces on IA-32 platform products.

### Signify (Philips Lighting)

*Hardware Engineer Intern*

Aug 2019 – Feb 2020

- Worked on a dual-mode self-adaptive LED/HID driver circuit for the EU market.

## TEACHING, MENTORING, AND SERVICE

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### University of Washington, Seattle

Jan 2024 – Present

*Graduate Teaching Assistant*

- Courses supported: EE538 Post-Tapeout Chip Test, EE478 Chip Testing, EE478 VLSI Capstone, EE477/525 VLSI-II, EE476 VLSI-I, and EE271 Digital Circuits.
- Led and mentored 5 TSMC180 tapeout projects.
- Developed a complete ASIC tapeout flow for TSMC 180 nm technology using the **Hammer** flow with custom TCL plugins.
- Implemented and maintained synthesis (Genus), place-and-route (Innovus), STA (Tempus), DRC/LVS (Calibre), padframe generation, and hierarchical flows with extracted timing models.

### University of Washington, Seattle

Winter 2025

*M.S. Admissions Triage Committee*

- Reviewed master's applicant materials and provided evaluation feedback.

### UW Semiconductor Society

Dec 2025 – Present

*Technical Member*

- Contributed to student community activities related to semiconductor design and systems research.

## EXPERTISE AND SKILLS

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- **Architecture and Modeling:** Strong background in AI hardware and CPU/GPGPU/FPGA/CGRA architecture; experienced with GPGPU-Sim/GPUWattch (Accel-Sim/AccelWattch) and gem5.
- **ASIC Design:** End-to-end ASIC design, tapeout, and mentoring experience in TSMC 180 nm and N28, including RTL design, verification, synthesis, place-and-route, STA, and signoff (DRC/LVS/PEX).
- **EDA Tools:** Cadence (Xcelium, Genus, Innovus, Tempus), Synopsys, and Mentor/Calibre.
- **Programming:** CUDA, Python, C++, and SystemVerilog; strong debugging skills and experience working with complex codebases.